Co-NP and the Asymmetry of NP

Asymmetry of NP

- **NP.** Decision problems for which there is a poly-time certifier
- **Ex.** SAT, HAM-CYCLE, COMPOSITES
- **Asymmetry of NP.** We only need to have short proofs of yes instances
- It may not be easy to show proofs of no instances
- **Ex 1.** SAT vs. TAUTOLOGY
  - Can prove a CNF formula is satisfiable by giving such an assignment
  - How could we prove that a formula is **not** satisfiable?
- **Ex 2.** HAM-CYCLE vs. NO-HAM-CYCLE
  - Can prove a graph is Hamiltonian by giving such a Hamiltonian cycle
  - How could we prove that a graph is not Hamiltonian?
- **Q.** How do we classify TAUTOLOGY?
  - SAT is NP-complete and SAT $\equiv_p$ TAUTOLOGY
  - But TAUTOLOGY is not even known to be in NP

NP and co-NP

- **NP.** Decision problem for which there is a poly-time certifier
- **Ex.** SAT, HAM-CYCLE, COMPOSITES

- **Def.** Given a decision problem X, its complement is $\overline{X}$: same problem with the yes and no answers reverse

- **Ex.** $\overline{X}$ {0, 1, 4, 6, 8, 9, 10, 12, 14, 15, …}

  X = {2, 3, 5, 7, 11, 13, 17, 23, 29, …}

- **Co-NP.** Complements of decision problems in NP
  - For no instance, there is a succinct disqualifier
- **Ex.** TAUTOLOGY, NO-HAM-CYCLE, PRIMES

NP = co-NP?

- **Fundamental question.** Does NP = co-NP?
  - Do yes instances have succinct certificates iff no instances do?
  - Consensus opinion: no

- **Theorem.** If NP ≠ co-NP, then P ≠ NP
- **Pf idea**
  - P is closed under complementation
  - If P = NP, then NP is closed under complementation
  - In other words, NP = co-NP
  - This is the contrapositive of the theorem

Good Characterizations

- **Good characterization.** [Edmonds 1965] NP ⋂ co-NP
  - If problem X is in both NP and co-NP, then:
    - For yes instance, there is a succinct certificate
    - For no instance, there is a succinct disqualifier
  - Provides conceptual leverage for reasoning about a problem

- **Ex.** Given a bipartite graph, is there a perfect matching
  - If yes, can exhibit a perfect matching
  - If no, can exhibit a set of nodes S such that $|N(S)| < |S|$

Good Characterizations

- **Observation.** $P \subseteq NP \cap co\text{-}NP$
- **Fundamental open question.** Does $P = NP \cap co\text{-}NP$?
  - Mixed opinions
  - Many examples where problem found to have a non-trivial good characterization, but only years later discovered to be in P
    - Linear programming [Khachiyan, 1979]
    - Primality testing [Agrawal-Kayal-Saxena, 2002]

- **Fact.** Factoring is in $NP \cap co\text{-}NP$, but not known to be in P

If poly-time algorithm for factoring, can break RSA cryptosystem

FACTOR is in NP ⋂ co-NP

- **FACTORIZE.** Given an integer x, find its prime factorization
- **FACTOR.** Given two integers x and y, does x have a nontrivial factor less than y?
- **Theorem.** FACTOR $\equiv_p$ FACTORIZE.
- **Pf.** Binary search to find a factor, divide the factor and repeat

- **Theorem.** FACTOR is in NP ∩ co-NP
- **Pf.**
  - Certificate: a factor p of x that is less than y
  - Disqualifier: the prime factorization of x (where each prime factor is ≥ y), along with a Pratt's certificate that each factor is prime

Primality Testing and Factoring

- **Q: Is PRIMES in P?**
  - Yes! (Agrawal-Kayal-Saxena, 2002)

- **Q: Is FACTOR in P?**
  - Consensus: No

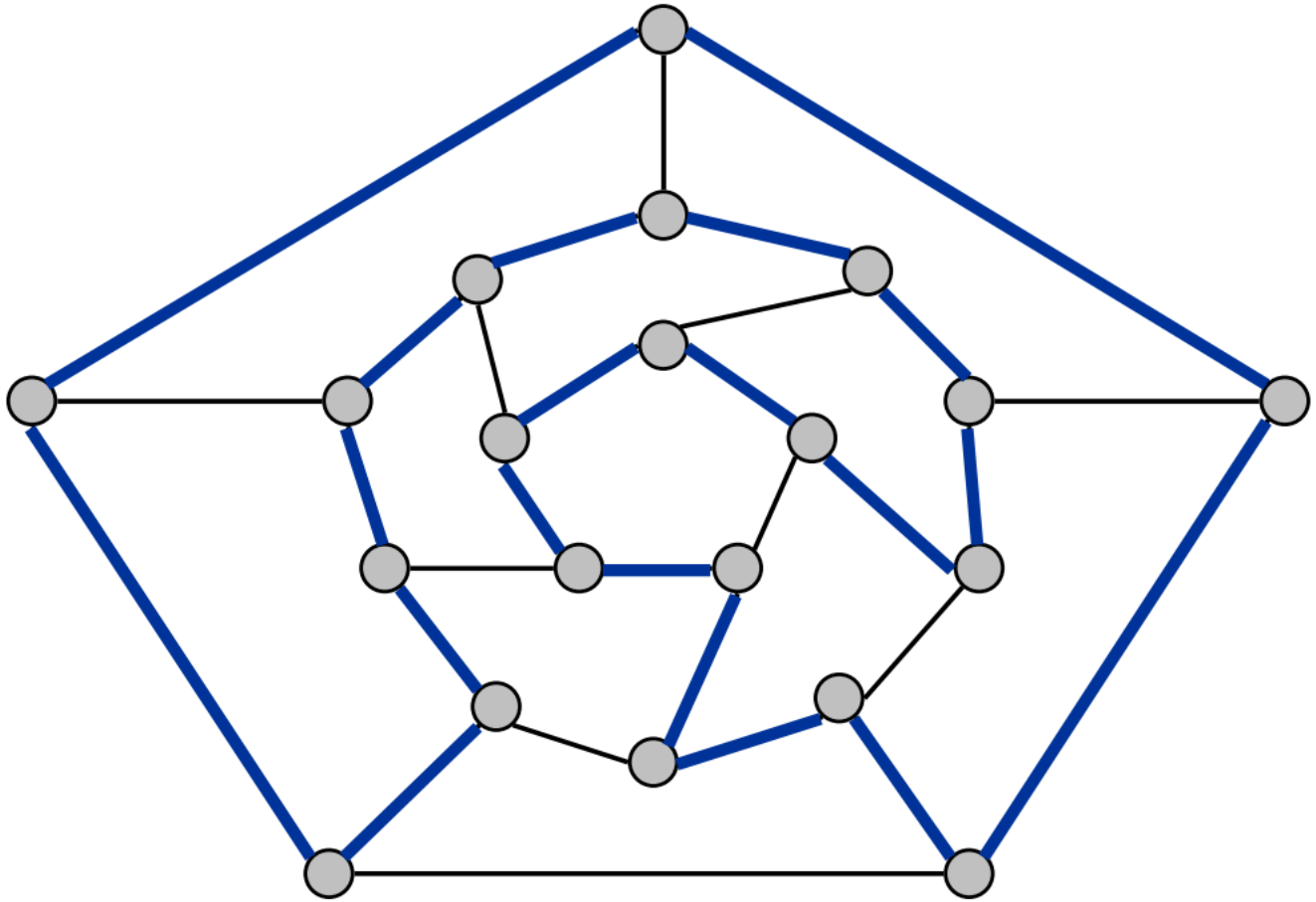
- **RSA cryptosystem**
  - Based on dichotomy between complexity of two problems
  - To use RSA, must generate large primes efficiently
  - To break RSA, must find efficient factoring algorithm

Sequencing Problems

- **Basic genres**
  - Packing problems: SET-PACKING, INDEPENDENT SET
  - Covering problems: SET-COVER, VERTEX-COVER
  - Constraint satisfaction problems: SAT, 3-SAT
  - **Sequencing problems: HAMILTONIAN-CYCLE, TSP**
  - Partitioning problems: 3D-MATCHING, 3-COLOR
  - Numerical problems: SUBSET-SUM, KNAPSACK

Hamiltonian Cycle

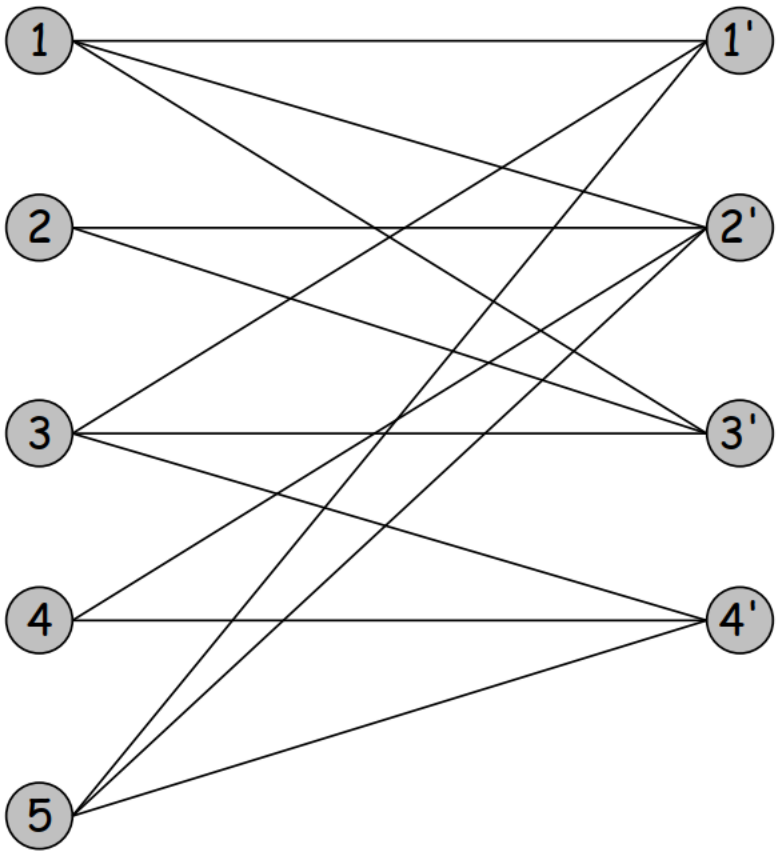
- **HAM-CYCLE:** given an undirected graph G = (V, E), does there exist a simple cycle that contains every node in V



YES: vertices and faces of a dodecahedron

Hamiltonian Cycle

- **HAM-CYCLE:** given an undirected graph G = (V, E), does there exist a simple cycle that contains every node in V

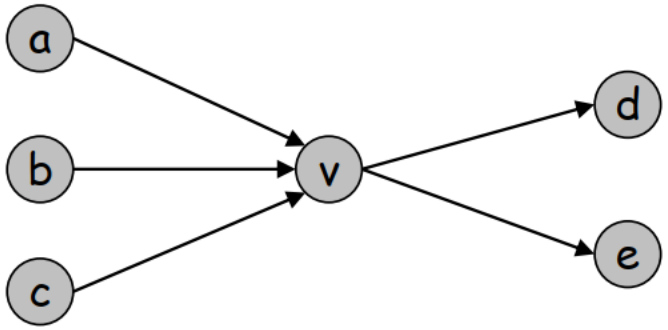


NO: bipartite graph with odd number of nodes

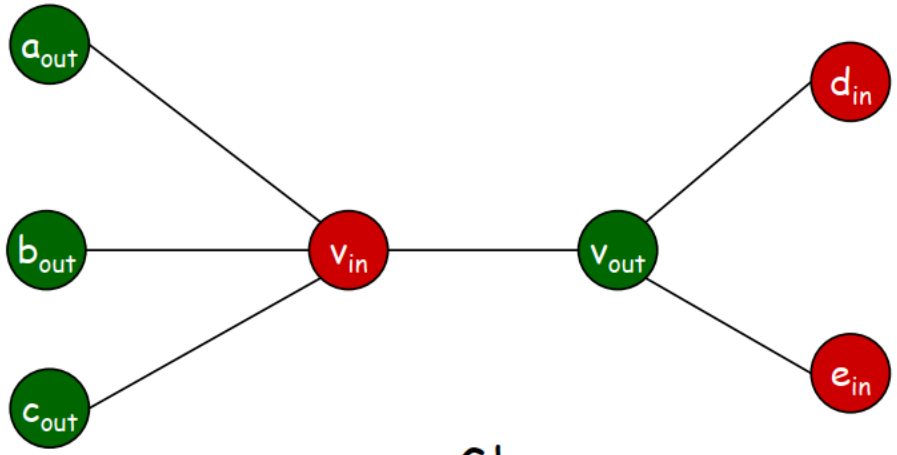
Directed Hamiltonian Cycle

- **DIR-HAM-CYCLE:** given a **digraph** G = (V, E), does there exists a simple directed cycle that contains every node in V?

- **Claim.** DIR-HAM-CYCLE $\leq_p$ HAM-CYCLE
- **Pf.** Given a directed graph G = (V, E), construct an undirected graph G'



G



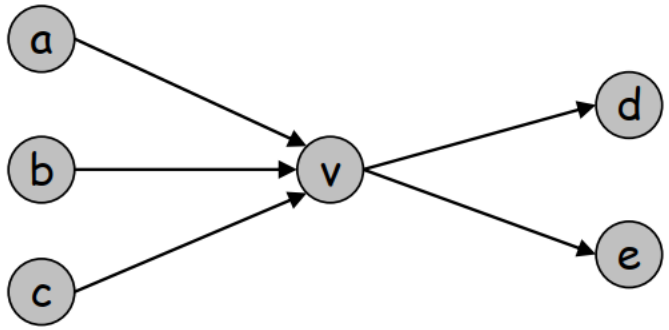
G'

Directed Hamiltonian Cycle

- **DIR-HAM-CYCLE:** given a **digraph** G = (V, E), does there exists a simple directed cycle that contains every node in V?

- **Recipe to establish NP-completeness of problem Y**
  - Step 1. Show that Y is in NP
  - Step 2. Choose an NP-complete problem X
  - Step 3. Prove that X $\leq_p$ Y

- **Claim.** DIR-HAM-CYCLE $\leq_p$ HAM-CYCLE
- **Pf.** Given a directed graph G = (V, E), construct an undirected graph G' with 3n nodes



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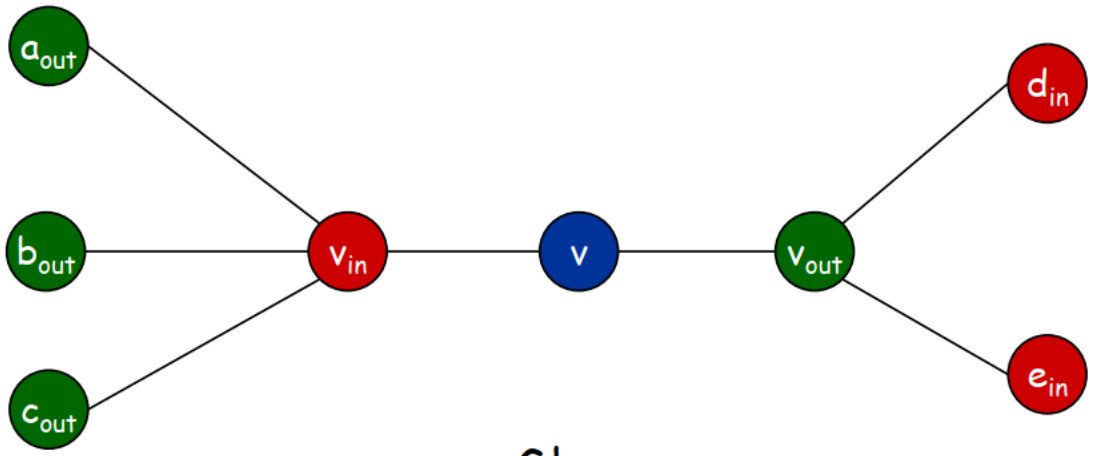


G'

Directed Hamiltonian Cycle

- **Claim.** G has a Hamiltonian cycle iff G' does

- **Pf. →**
  - Suppose G has a directed Hamiltonian cycle
  - Then G' has an undirected Hamiltonian cycle (same order)

- **Pf. ←**
  - Suppose G' has an undirected Hamiltonian cycle
  - The cycle must visit nodes in G' using one of following two orders:
  - …, B, G, R, B, G, R, B, G, R, B, …
  - …, B, R, G, B, R, G, B, R, G, B, …
  - Blue nodes in this cycle make up directed Hamiltonian cycle in G, or reverse of one

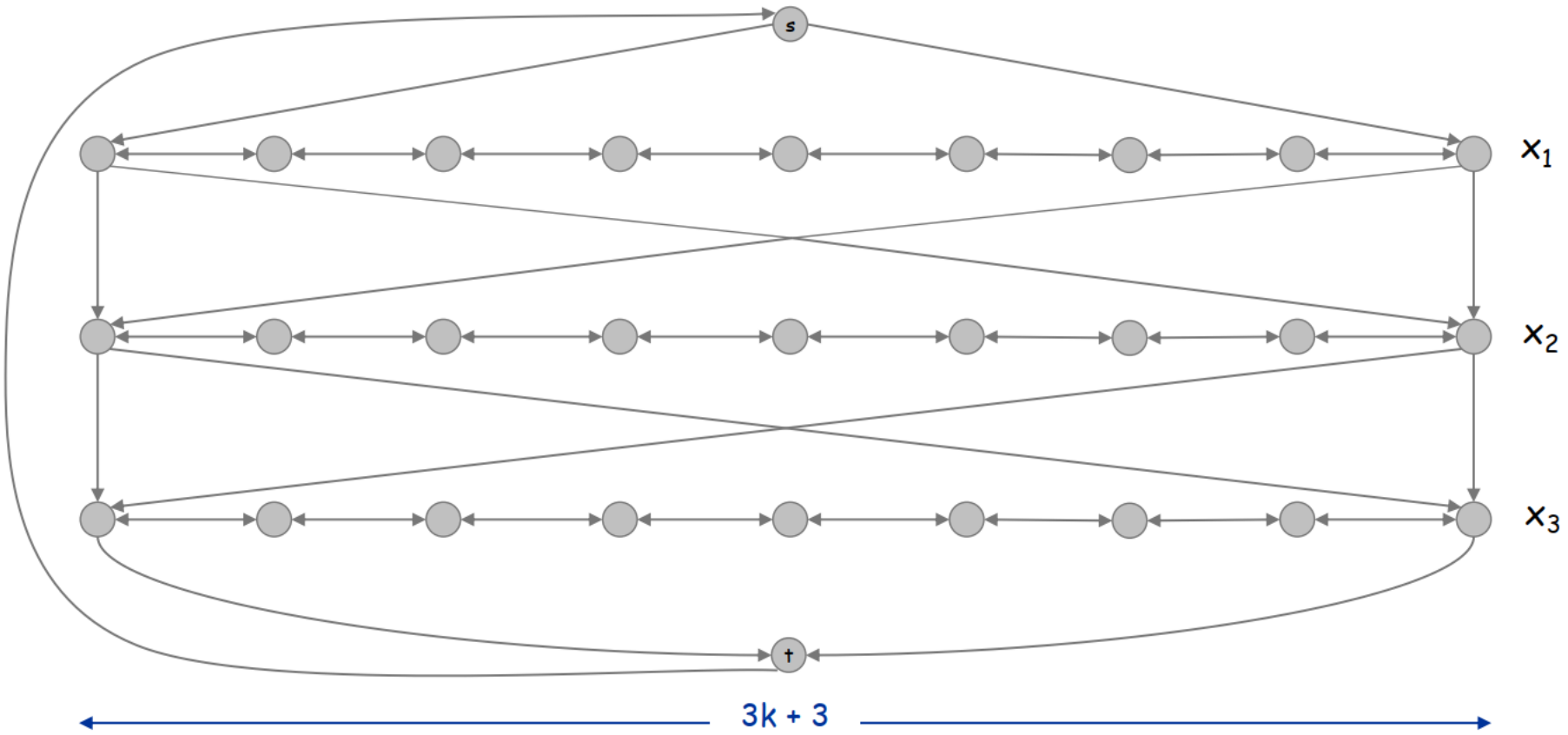
3-SAT Reduces to Directed Hamiltonian Cycle

- **Claim.** 3-SAT $\leq_p$ DIR-HAM-CYCLE

- **Pf.** Given an instance ϕ of 3-SAT, we construct an instance of DIR-HAM-CYCLE that has a Hamiltonian cycle iff ϕ is satisfiable

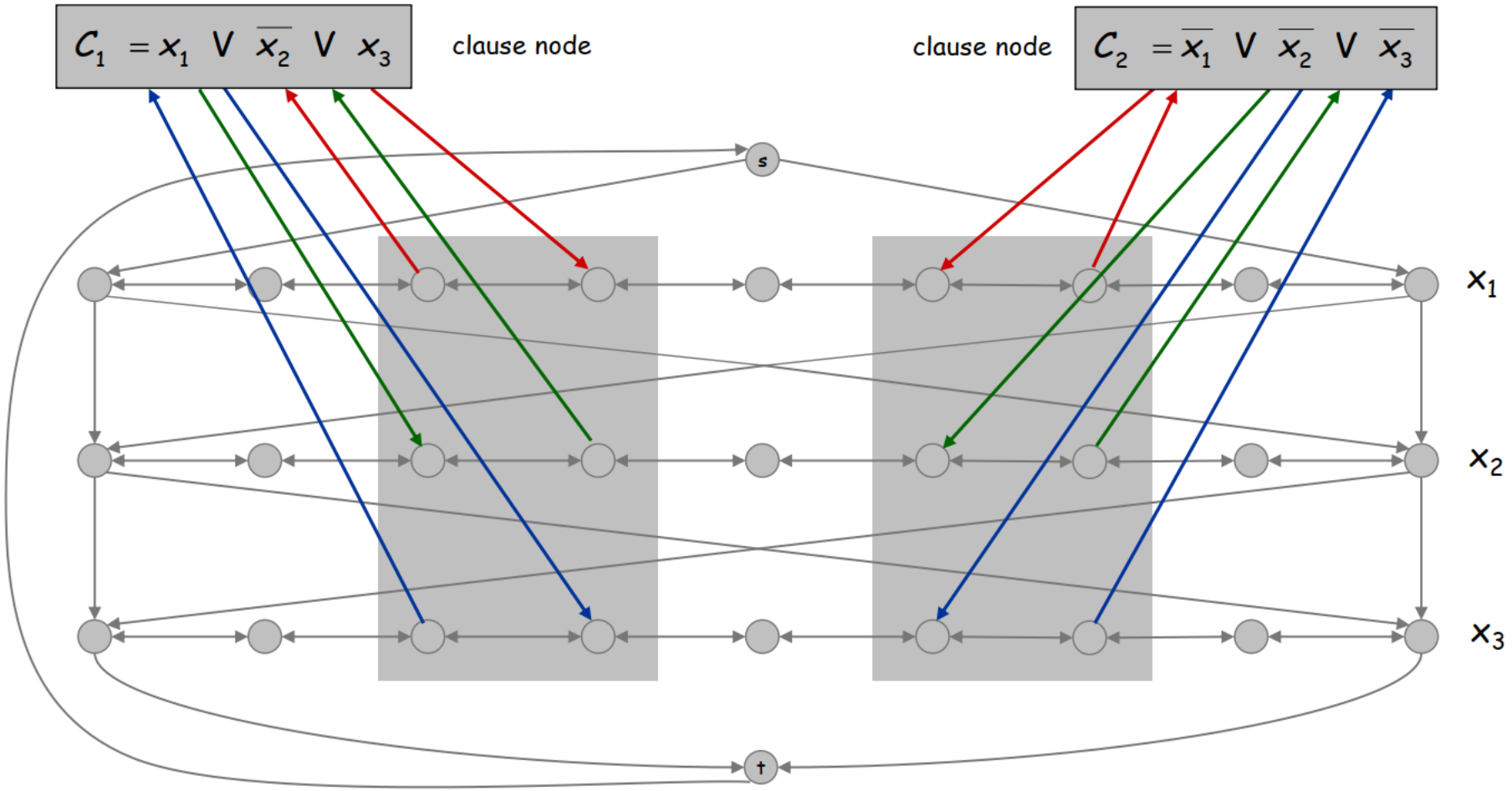
3-SAT Reduces to Directed Hamiltonian Cycle

- **Construction. Given 3-SAT instance ϕ** with n variable x_i and k clauses.
  - Construct G to have 2^n Hamiltonian cycles
  - Intuition: traverse path i from left to right $\leftrightarrow$ set variable $x_i = 1$



3-SAT Reduces to Directed Hamiltonian Cycle

- **Construction. Given 3-SAT instance ϕ** with n variable x_i and k clauses.
  - For each clause: add a node and 6 edges

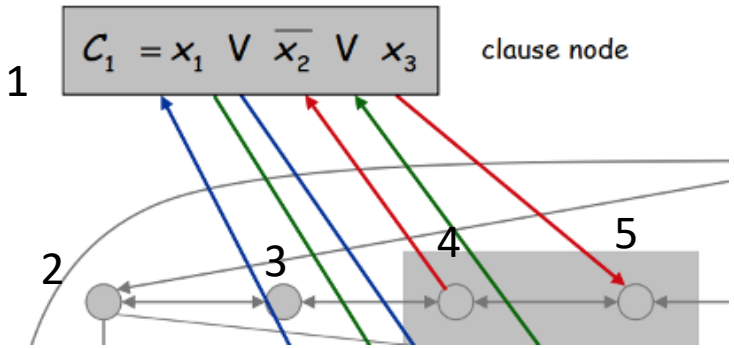


3-SAT Reduces to Directed Hamiltonian Cycle

- **Claim.** ϕ is satisfiable iff G has a Hamiltonian cycle
- **Pf. →**
  - Suppose 3-SAT instance has satisfying assignment x^*
  - Then, define Hamiltonian cycle in G as follows:
    - If $x^*_i = 1$, traverse row i from left to right
    - If $x^*_i = 0$, traverse row i from right to left
    - For each clause C_j, there will be at least one row i in which we are going in "correct" direction to splice node C_j into tour

3-SAT Reduces to Directed Hamiltonian Cycle

- **Claim.** ϕ is satisfiable iff G has a Hamiltonian cycle
- **Pf. ←**
  - Suppose G has a Hamiltonian cycle T
  - If the cycle T enters clause node C_j, it must depart on mate edge
    - Thus, nodes immediately before and after C_j are connected by an edge e in G
    - Removing C_j from cycle, and replacing it with edge e yields Hamiltonian cycle on G – { C_j }
    - Continuing in this way, we are left with Hamiltonian cycle T' in G – { C_1, C_2, …, C_k}
    - Set $x^*_i = 1$ iff T' traverse row i left to right
    - Since T' visits each clause node C_j, at least one of the paths is traversed in "correct" direction, and each clause is satisfied



Next Time:
NP and Computational Intractability (Cont.)